



NALLA NARASIMHA REDDY

Education Society's Group of Institutions - Integrated Campus

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ABSTRACT

Enhancing Predictive Accuracy for Agricultural Crop Yields in Indian States

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Agricultural productivity is critical for the sustenance and economic growth of India, a country with diverse climatic regions and crop practices. Accurate prediction of crop yields is essential for optimizing agricultural outputs, ensuring food security, and planning resource allocation. This study introduces an innovative approach by integrating power transformations, within machine learning models to enhance the predictive accuracy of agricultural crop yields in various Indian states. We collected a comprehensive dataset comprising climatic variables, soil properties, crop types, and historical yield data across different Indian states. We evaluated several machine learning models, including Linear Regression, Random Forest, and Gradient Boosting, to establish a benchmark for comparison. The Gradient Boosting model, post-transformation, exhibited the highest accuracy, underscoring the potential of combining power transformations with ensemble learning techniques for crop yield prediction. This approach provides insights into the relative importance of various factors affecting crop yields. The study emphasizes the role of advanced data preprocessing techniques, such as power transformations, in unlocking the full potential of machine learning for agricultural applications. By enhancing the predictive accuracy of crop yield models, stakeholders can make more informed decisions, leading to improved agricultural productivity and sustainability in the diverse and challenging landscape of Indian agriculture. explain this project.

Keywords: Crop yield prediction, Machine learning, Crop management, Soil properties